

Design and Performance Analysis of Slotted Patch Antenna for Sub-6 GHz 5G Communications

Md. Najmul Hossain
Department of Electrical, Electronic and Communication
Engineering
Pabna University of Science and Technology
Pabna-6600, Bangladesh
najmul_eece@pust.ac.bd

Md. Sarwar Hosain
Department of Information and Communication Engineering
Pabna University of Science and Technology
Pabna-6600, Bangladesh
sarwar.ice@pust.ac.bd

Sayed Shifat Ahmed
Department of Electrical and Electronic Engineering
RTM Al-Kabir Technical University
Sylhet-3100, Bangladesh
shikor99@gmail.com

Tetsuya Shimamura
Graduate School of Science and Engineering
Saitama University
Saitama 338-8570, Japan
shima@mail.saitama-u.ac.jp

Abstract— Fifth-generation (5G) technology has brought a revolutionary change in the field of telecommunication by offering sufficient bandwidth, higher data transfer rates, huge network capacity, and so on. To augment this legacy, an antenna plays an essential part in any wireless network. Therefore, in this article, a high-gain slotted patch antenna (SPA) has been designed and suggested for a Sub-6 GHz band 5G application. It is capable of operating at 3.6282 GHz in the 5G N77 band, as well as in Sub-6 GHz 5G, LTE Band 42, and WiMAX. The material, namely Rogers RT5880 (dielectric constant = 2.2, tangent loss = 0.0009), is used in the substrate layer to be operated, whose dimension is optimized as 38x38x1.575 mm³. The proposed antenna is simulated and then analyzed by using CST simulation tools. It exhibits an excellent performance regarding reflection coefficient ($|S_{11}|$ = -55.459 dB) at the 3.6282 GHz frequency with an approximate unity VSWR (1.0035 $\approx$ 1). Also, it offers an impedance bandwidth of 165.3 MHz, with a maximum radiation gain of 6.836 dB and directivity of 7.068 dBi at the resonance frequency. The radiation efficiency of the suggested antenna is 94.795%. Simulation results verify that the proposed antenna exhibits high radiation gain, higher radiation efficiency, and excellent impedance bandwidth.

Keywords—5G Technology, Sub-6 GHz, CST, Slotted Patch Antenna, High Gain.

I. INTRODUCTION

Advancements in fifth-generation (5G) wireless communication systems have enabled superior quality of service (QoS) and significantly faster data transfer capabilities compared to previous generations, massive connectivity, more secure networks, low latency, and many more. It has ushered in a new era for various sectors, including automotive, transportation systems, energy, virtual reality (VR), education, healthcare, entertainment, and manufacturing industries [1-2]. In November 2015, the World Radio Communications Conference (WRC) agreed to allocate (3.4 - 3.6) GHz frequencies for 5G communication systems [3]. On the other hand, the Federal Communications Commission (FCC) has divided the 5G spectrum into four bands. Among them, the Sub-6 GHz frequency band has increased the interest of RF researchers because it allows long-distance communication at faster data transfer rates with “zero” latency (delay less than 1 ms). Table I exhibits the Sub-6 GHz spectrum for the worldwide 5G

communication systems, indicating that almost all countries consider this frequency band for 5G technologies [4]. Microstrip patch antennas (MPAs) are very popular among RF researchers and are widely used in various technologies, such as WLAN, WiFi, 3G, LTE, 5G, and WiMAX applications, due to their lightweight, cost-effective, low-profile design with sustainable gain and improved bandwidth.

Different techniques are used to enhance the antenna's performance, such as slot formation, defective ground structure (DGS), array structure, meta-material structure, and so on [5-6]. In [7], a circular patch antenna employing an artificial dielectric substrate for Sub-6 GHz 5G communications achieves 179.9 MHz bandwidth and 6.391 dB gain at 3.5 GHz, though its dimension is large (45x40x4.5 mm³). A rectangular rhombic-slotted patch antenna is reported in [8] with 47.46x36.57x1.5 mm³ size, offering 6.158 dBi gain at 3.5305 GHz but limited bandwidth (116.4 MHz). An elliptical patch in [9] provides 5.01 dB gain with 25.2x48 mm² area at 3.5 GHz. In [10], a dual-band slotted antenna on Rogers 5880 (50x30x1.57 mm³) exhibits 7.52 dBi and 5 dBi gain but only 50 MHz bandwidth in both bands. A dual-band H-slotted design [11] achieves low gains (2.18 dB at 3.76 GHz, 2.75 dB at 6.1 GHz). Patch antennas using multiple substrates are studied in [12], [13], where Arlon AD300C shows superior performance (7.15 dBi at 5.65 GHz, 52.92x55.56x1.2 mm³). A high-gain array in [14] demonstrates 6.77 dBi at 3.51 GHz but with larger dimensions (60x70x1.6 mm³).

TABLE I. THE SPECTRUM OF GLOBAL 5G TECHNOLOGY [4]

Countries	(3 – 4) GHz Band	(4 – 5) GHz Band	(5 – 7) GHz Band
China	(3.3 – 3.6) GHz	(4.5 – 5.0) GHz	–
UK	(3.4 – 3.8) GHz	–	–
USA	(3.45 – 3.7) GHz (3.7 – 3.98) GHz	(4.49 – 4.99) GHz	(5.9 – 7.1) GHz
Canada	(3.47 – 3.65) GHz (3.65 – 4.0) GHz	–	(5.9 – 7.1) GHz
Australia	(3.4 – 3.7) GHz	–	–
Italy	(3.6 – 3.8) GHz	–	–
India	(3.4 – 3.6) GHz	–	–
Malaysia	3.5 GHz	–	–
Korea	(3.4 – 3.7) GHz (3.7 – 4.0) GHz	–	–
Japan	(3.6 – 4.1) GHz	(4.5 – 4.9) GHz	–
EU	(3.4 – 3.8) GHz	–	(5.9 – 6.4) GHz

This paper proposes a compact slotted patch antenna ($38 \times 38 \times 1.575$ mm³) for Sub-6 GHz 5G services (3.5101–3.6754 GHz). The antenna achieves superior performance with a reflection coefficient of -55.459 dB, radiation gain of 6.836 dB, efficiency of 94.795% , and an impedance bandwidth of 165.3 MHz. Owing to its compact size and excellent radiation characteristics, it is a strong candidate for 5G applications. Section II details the antenna design, Section III presents simulation results, and Section IV concludes the work.

II. PROPOSED SLOTTED PATCH ANTENNA DESIGN

The proposed slotted patch antenna, designed and analyzed in CST Studio, consists of a modified square patch, Rogers RT5880 substrate ($\epsilon_r = 2.2$, thickness = 1.575 mm), and copper ground plane (0.02 mm). The substrate dimensions are $L_s = W_s = 38$ mm, identical to the ground plane. Two identical rectangular slots (13.4×2 mm²) are etched on the patch sides, while four rectangular and F-shaped slots are introduced at the top and bottom sections. A 50Ω coaxial probe feed is positioned 3.7 mm offset from both axes. Geometrical details are presented in Fig. 1 and Table II.

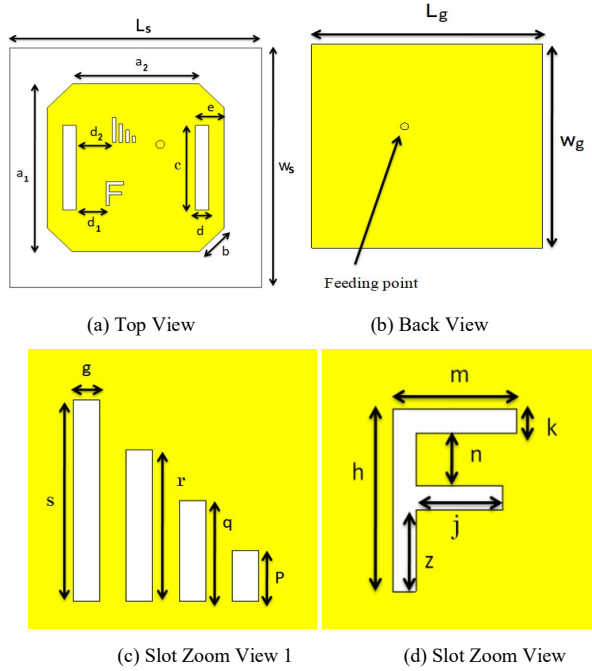


Fig. 1. Geometry of the proposed slotted patch antenna (SPA).

TABLE II. DIMENSION DETAILS OF THE SUGGESTED SPA

Parameters	Value(mm)	Parameters	Value(mm)
L_s	38	W_s	38
L_g	38	W_g	38
t_s	1.575	t_g	0.02
t_p	0.02	p	1
a_1	26.60	q	2
a_2	19.13	r	3
b	5.28	s	4
c	13.40	g	0.5
d	2	h	3.8
d_1	4.5	k	0.5
d_2	5.5	m	2.7
e	4.3	n	1.1
j	1.9	z	1.7

III. SIMULATION RESULTS AND ANALYSIS

This section has carried out all the simulations and parametric studies of the proposed slotted high-gain patch antenna. Fig. 2 depicts the return loss ($|S_{11}|$) characteristics, where the antenna resonates at 3.6282 GHz with a remarkably low return loss of -55.459 dB. The corresponding impedance bandwidth, defined at the -10 dB threshold, is 165.3 MHz (ranging from 3.5101 GHz to 3.6754 GHz), which adequately covers a significant portion of the Sub-6 GHz 5G spectrum. Furthermore, Fig. 3 demonstrates an excellent VSWR of approximately unity ($1.0035 \approx 1$) at the resonance frequency, ensuring superior impedance matching and minimal reflection losses. These results indicate that the proposed SPA not only provides stable resonance but also achieves efficient power transfer.

The simulated radiation characteristics of the proposed slotted patch antenna (SPA) are shown in Fig. 4–6. At 3.6282 GHz, the SPA achieves a maximum realized gain of 6.836 dB, directivity of 7.068 dBi, and high efficiency (94.795%), indicating minimal losses. The 1D gain and directivity responses versus frequency (Fig. 4c) exhibit close agreement, confirming high radiation efficiency. The 2D polar gain pattern at $\phi = 90^\circ$ (Fig. 5) shows a main lobe magnitude of 6.84 dB with 1° direction and 84.5° angular width, demonstrating stable and directional radiation characteristics ensure enhanced link quality, making the proposed SPA a strong candidate for compact, high-performance antenna solutions in Sub-6 GHz 5G communication systems.

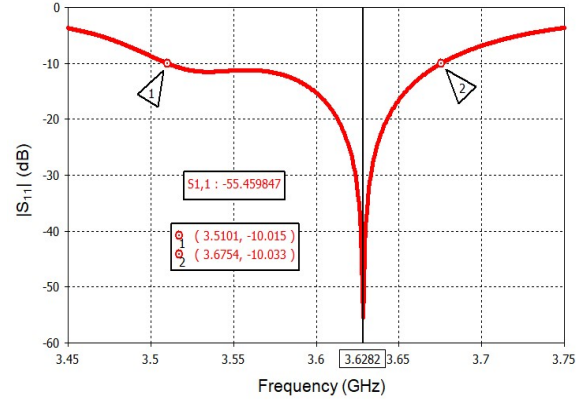


Fig. 2. Return loss ($|S_{11}|$) analysis of the proposed SPA model

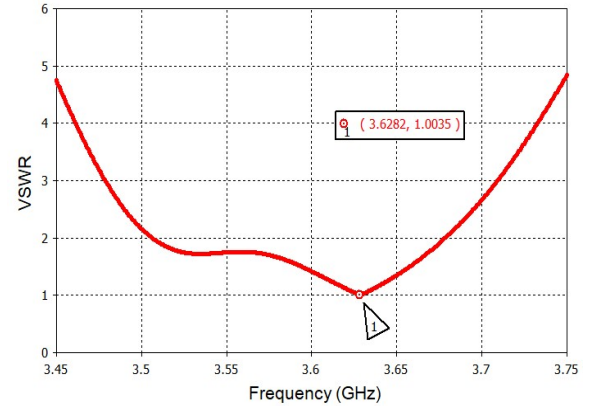
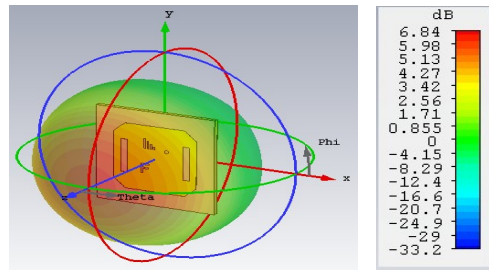
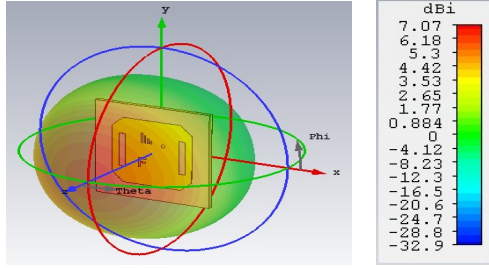


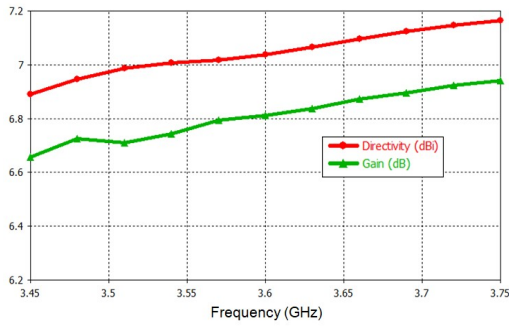
Fig. 3. VSWR analysis of the proposed SPA model.



(a) 3D radiation gain



(b) 3D directivity



(c) Linear plot of gain vs directivity of the proposed SPA model
Fig. 4. Gain and directivity of the proposed SPA at 3.6282 GHz.

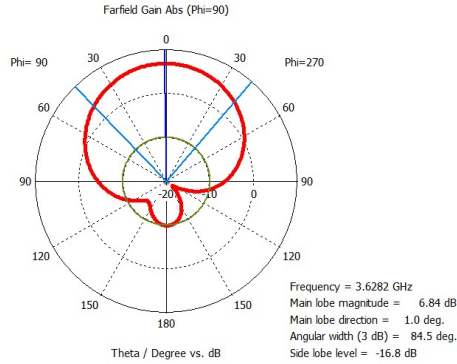


Fig. 5. Far-field (polar) Gain pattern curve at 3.6282 GHz when $\phi=90^\circ$.

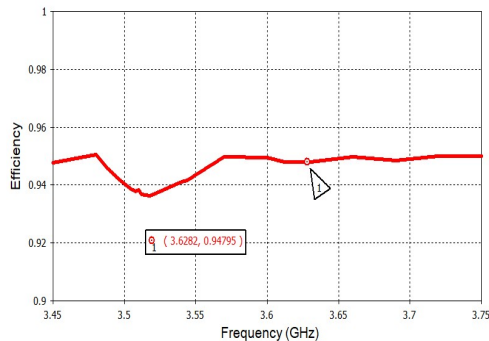
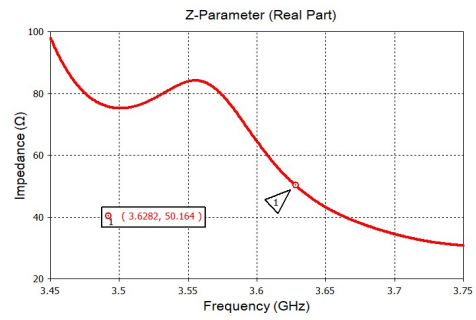
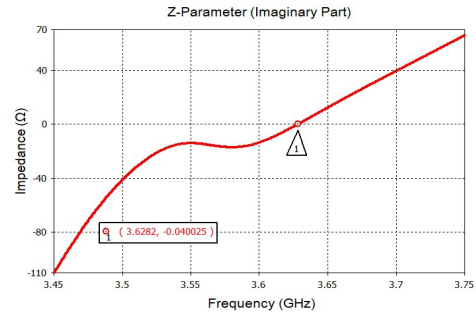


Fig. 6. Radiation efficiency vs. frequency plot of the suggested SPA.



(a) Z-Parameter (Real Part)



(b) Z-Parameter (Imaginary Part)

Fig. 7. Impedance profile (Z-Parameter) of the suggested SPA.

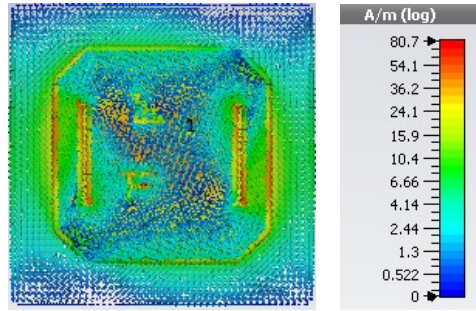


Fig. 8. Distribution of surface current at 3.6282 GHz.

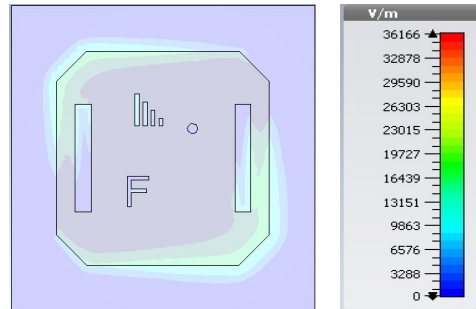


Fig. 9. E-field distribution of the suggested SPA at 3.6282 GHz.

The impedance profile of the proposed SPA is $(50.164 - j0.04) \Omega$ at 3.6282 GHz, nearly ideal for 50Ω systems with negligible reactance, as shown in Fig. 7. This precise matching yields excellent return loss (-55.459 dB) and unity-like VSWR (1.0035). The surface current distribution (Fig. 8) reaches 80.71 A/m, confirming efficient excitation of the dominant mode and contributing to the realized gain (6.836 dB) and directivity (7.068 dBi). The E-field distribution (Fig. 9) peaks at 36,166 V/m, ensuring strong radiation with minimal loss. These characteristics validate the SPA's efficiency and suitability for Sub-6 GHz 5G communication systems.

IV. COMPARATIVE ANALYSIS

A comparison of the suggested SPA model with previously published works is presented in Table III, which reports the model's size, frequency, bandwidth, and gain. The proposed antenna achieves a compact size of $38 \times 38 \times 1.575$ mm³, smaller than most references [7], [8], [12], [13], and [14], making it suitable for space-constrained applications. It resonates at 3.6282 GHz, comparable with [7], [8], [12], and [14]. The bandwidth of 165.3 MHz improves over [8] and [12], while maintaining a balance with size. With a gain of 6.836 dB, it surpasses [12] and matches [7] and [14], demonstrating an efficient trade-off between compactness, bandwidth, and gain for modern wireless communication systems.

TABLE III. COMPARATIVE STUDY

Ref.	Size (mm ³)	Operating Freq. (GHz)	Resonance Freq. (GHz)	BW (MHz)	Gain (dB)
[7]	50×53×4.5	3.4056 – 3.5855	3.50	179.9	6.391
[8]	47.46×36.57×1.5	3.5887 – 3.7051	3.5305	116.4	Not reported
[12]	47×31.698×1.575	3.4351 – 3.5648	3.50	129.7	4.660
[13]	52.92×55.56×1.2	5.615 – 5.75	5.65	135	7.15
[14]	60×70×1.6	3.46 – 3.56	3.51	~100	6.77
This Work	38×38×1.575	3.510 – 3.675	3.6282	165.3	6.836

V. CONCLUSION

In this paper, a coaxial-fed slotted patch antenna (SPA) with an overall dimension of $38 \times 38 \times 1.575$ mm³ is suggested for the Sub-6 GHz 5G communication services. It can operate at 3.6282 GHz in the 5G N77 band, LTE Band 42, Sub-6 GHz 5G, and WiMAX. The special features of the suggested slotted antenna are its compact size, wide impedance bandwidth, high radiation gain, proper impedance matching, and high radiation efficiency. It is designed on the Rogers RT5880 material and operates at a frequency of 3.6282 GHz with an excellent reflection coefficient ($|S_{11}| = -55.459$ dB) and approximately unity VSWR. It offers an impedance bandwidth of 165.3 MHz with a radiation efficiency of 94.795% and a radiation gain of 6.836 dB. Also, its impedance profile is $(50.164 - j0.040) \Omega$, which denotes 50Ω pure resistivity. Since the suggested SPA exhibits high radiation gain, higher radiation efficiency, and excellent impedance bandwidth, it can be considered a suitable candidate to effectively apply for 5G (Sub-6 GHz band) communications.

REFERENCES

- [1] H. Th. Alrikabi, A. H. M. Alaidi, A. S. Abdalrada, and F. T. Abed, "Analysis the Efficient Energy Prediction for 5G wireless communication Technologies," *International Journal of Emerging Technologies in Learning (IJET)*, vol. 14, no. 08, p. 23, Apr. 2019, doi: 10.3991/ijet.v14i08.10485.
- [2] M. M. Rahman and H. -G. Ryu, "Design of Ultra-Wideband Patch Antenna for Multiple Communication Services," *2022 International Conference on Electronics, Information, and Communication (ICEIC)*, Jeju, Korea, Republic of, 2022, pp. 1–3, doi: 10.1109/ICEIC54506.2022.9748692.
- [3] Ullah, S. Ullah, R. Ullah, F. Faisal, I. B. Mabrouk and M. J. A. Hasan, "A 10-Ports MIMO Antenna System for 5G Smart-Phone Applications," in *IEEE Access*, vol. 8, pp. 218477–218488, 2020, doi: 10.1109/ACCESS.2020.3042750.
- [4] Hasan, M., Islam, M. T., Samsuzzaman, M., Baharuddin, M. et al., "Md. M. Hasan et al., "Gain and isolation enhancement of a wideband MIMO antenna using metasurface for 5G sub-6 GHz communication systems," *Scientific Reports*, vol. 12, no. 1, Jun. 2022, doi: 10.1038/s41598-022-13522-5.
- [5] Badr, S., & Hamad, E. K., "Design of multiband microstrip patch antenna for WiMax, C-band and X-band applications," *Aswan Engineering Journal*, pp.1–7, 2018.
- [6] M. M. Rahman, H. -G. Ryu and M. N. Hossain, "A Compact Slotted Patch Antenna Design for Multiband Applications," *2021 IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob)*, Bandung, Indonesia, 2021, pp. 147–150, doi: 10.1109/APWiMob51111.2021.9435241.
- [7] Y. N. Rahmawati, H. Ludiati, and N. Sardjito, "The characteristic of a 3.5 GHz circular patch antenna using Open-Ring artificial dielectric," *Advances in Engineering Research/Advances in Engineering Research*, Jan. 2021, doi: 10.2991/aer.k.211106.062.
- [8] B. G. Hakanoglu, O. Sen, B. Koc, S. E. Hayber and M. Turkmen, "Defected Grounded Rectangular Patch Antenna with Rhombic-Shaped Slots for Early Phase 5G Applications," *2020 XXXIIIrd General Assembly and Scientific Symposium of the International Union of Radio Science*, Rome, Italy, 2020, pp. 1–4, doi: 10.23919/URSIGASS49373.2020.9232027.
- [9] N. Ferdous, G. C. Hock, S. H. AHamid, M. N. A. Raman, T. S. Kiong, and M. Ismail, "DESIGN OF A SMALL PATCH ANTENNA AT 3.5 GHZ FOR 5G APPLICATION," *IOP Conference Series Earth and Environmental Science*, vol. 268, no. 1, p. 012152, Jun. 2019, doi: 10.1088/1755-1315/268/1/012152.
- [10] Q. Amjad, A. Kamran, F. Tariq and R. Karim, "Design and Characterization of a Slot based Patch Antenna for Sub-6 GHz 5G Applications," *2019 Second International Conference on Latest trends in Electrical Engineering and Computing Technologies (INTELLECT)*, Karachi, Pakistan, 2019, pp. 1–6, doi: 10.1109/INTELLECT47034.2019.8955475.
- [11] M. F. Nakmouche, A. M. M. A. Allam, D. E. Fawzy, D. Bing Lin and M. F. Abo Sree, "Development of H-Slotted DGS Based Dual Band Antenna Using ANN for 5G Applications," *2021 15th European Conference on Antennas and Propagation (EuCAP)*, Dusseldorf, Germany, 2021, pp. 1–5, doi: 10.23919/EuCAP51087.2021.9411213.
- [12] N. Ramli, S. Khan, T. Khalifa, and N. H., "Design and Performance Analysis of Different Dielectric Substrate based Microstrip Patch Antenna for 5G Applications," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 8, Jan. 2020, doi: 10.14569/ijacsa.2020.0110811.
- [13] B. Tütüncü and M. Kösem, "Substrate analysis on the design of Wide-Band antenna for sub-6 GHz 5G communication," *Wireless Personal Communications*, vol. 125, no. 2, pp. 1523–1535, Mar. 2022, doi: 10.1007/s11277-022-09619-9.
- [14] R. Song et al., "High-conductive graphene film based antenna array for 5G mobile communications," *International Journal of RF and Microwave Computer-Aided Engineering*, vol. 29, no. 6, p. e21692, Jan. 2019, doi: 10.1002/mmce.21692.